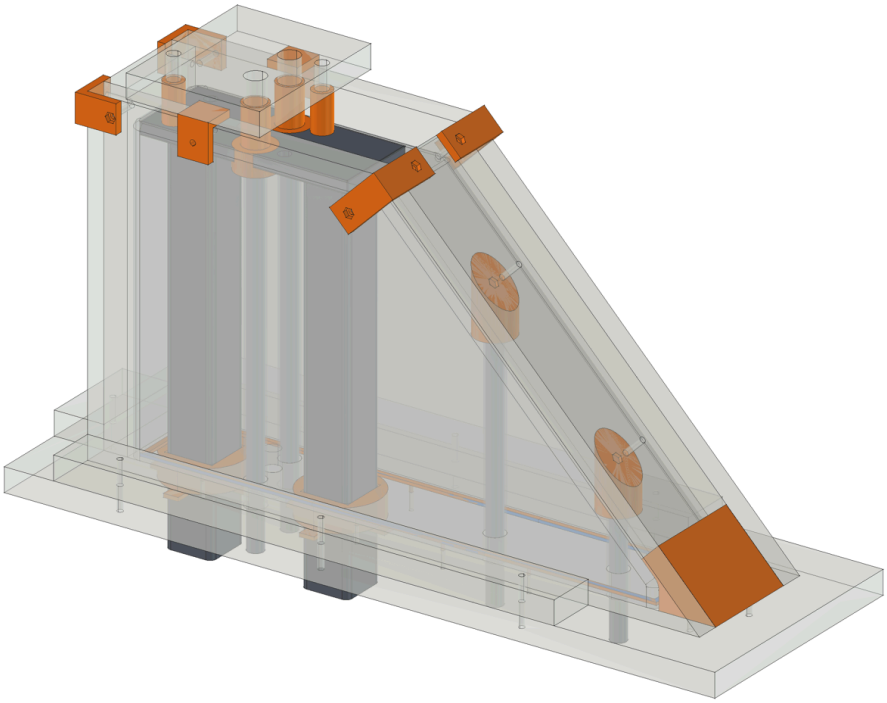
Upright molds

DIY CNC epoxy-granite mill · October 2026

How to make, put together and take apart the mold for the two uprights (two identical molds, cast together). The first section (how the molds work) and the last three (pouring, taking it apart, fixes) are the same in every mold guide; molds-complete.pdf has all the molds and the sheet layout for the boards.

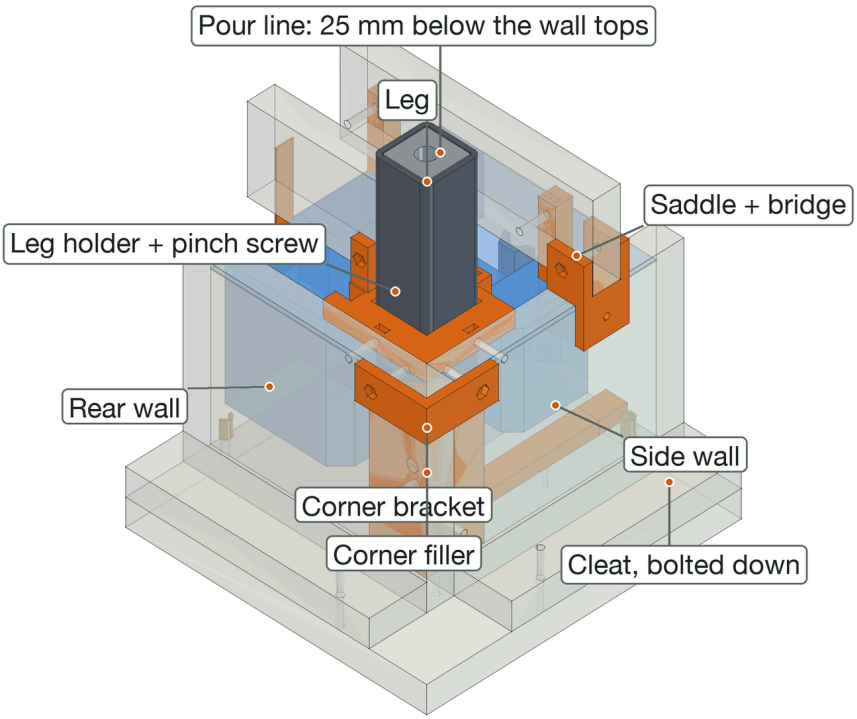
The build guide covers the casting itself (mix, pour, cure); this guide is everything before and after that. All the sizes and hole positions come from the model (`model/build_cnc.py`), so if you change the design, re-run it and the guides, hole maps and printed parts all follow.



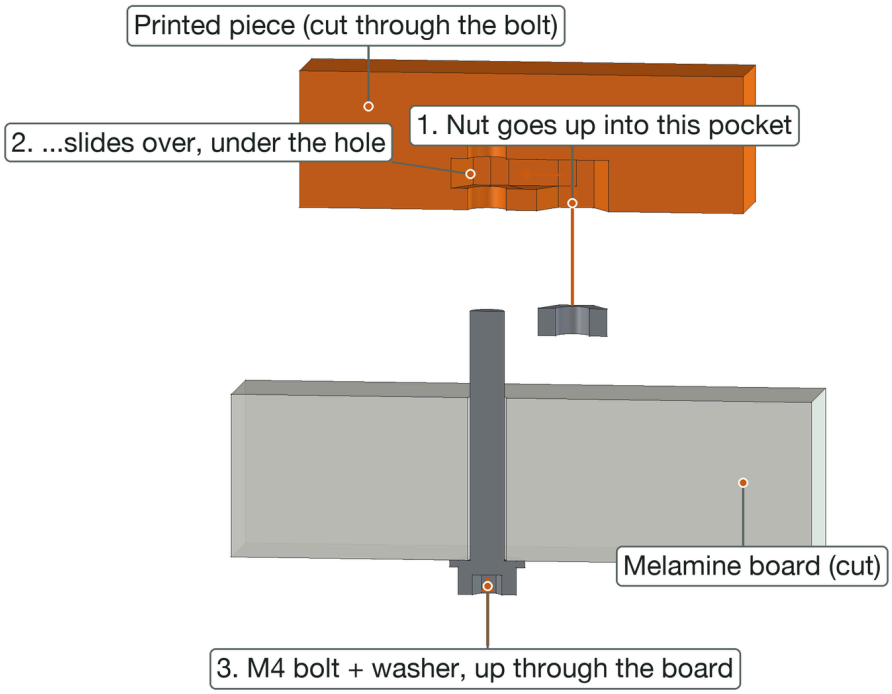
The order of work: print the kit (start first: it is days of printing) → cut and drill the boards → cut the foam → wax → assemble the mold → cast → take it apart. The castings go in this order: base rear half, base front half (against the rear half), the two uprights together, the beam.

1. How the molds work

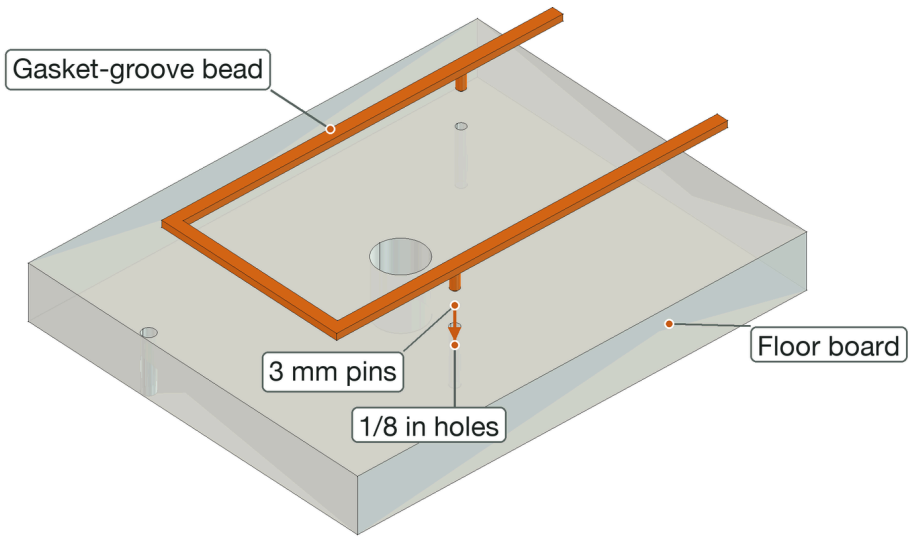
Walls stand 25 mm above the pour line. All the bolts that hold the box together go through that band, so none passes through a face that gets cast against. Fill to the line, not to the top. Strips of board (cleats) bolted down along the outside of each wall's foot keep the walls from moving out; printed corner brackets hold the corners square at the top.



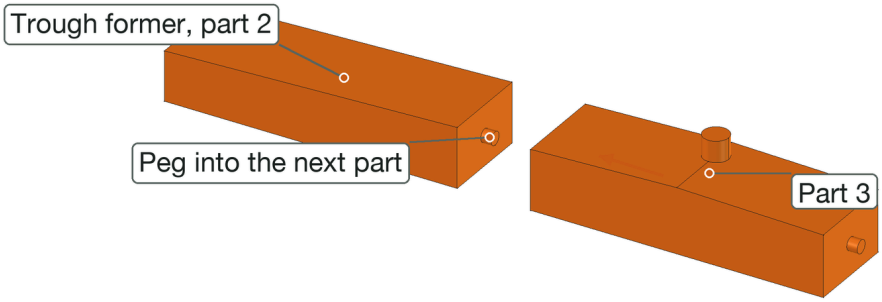
Printed pieces bolt on from outside. Each piece that forms part of the casting's surface (the formers and fillers) has a nut built into it: the nut drops into a pocket in the face that sits against the board, slides sideways under the bolt hole, and an M4 bolt (M3 on thin pieces) goes through the board from outside into it. The pocket is on the board side, so nothing opens onto a cast face, and the bolt comes out from outside after casting. Fit the nuts before the pieces go on.



Thin pieces have pins. The gasket-groove beads and the motor relief are too thin for a nut; their 3 mm pins push into 1/8 in holes. The casting presses them onto the board.



Pieces longer than the printer bed are printed in parts with a peg on one cut face and a socket in the other. Push them together; each part has its own bolts to the board, so they need no glue.



Fits are 0.15 mm a side: snug, but they go together by hand. Board thickness matters: the saddles and cradles are drawn for 19.05 mm board (Home Depot's 3/4 in white melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

Material	Amount	Notes
Melamine board, 19 mm (3/4 in)	about 1.0 m2	all the molds come out of two 4 x 8 ft sheets (layout in molds-complete.pdf)
PETG	0.31 kg	this mold's printed parts (sliced); all the molds and machine parts fit in two 1 kg rolls
Paste wax (Partall #2) and PVA	1 each	release, 3-5 coats
3/8 in quarter-round molding (PVC)	a few meters	the 10 mm outside edges of the castings
2x4 offcut	4 blocks per upright mold	52 mm long

Hardware (zinc-plated is fine; grease anything that touches the mix). Buy about 10 % spare of the small bolts, nuts and washers.

Item	Qty
M4 nut	42
M4 washer	42
M4 x 30 bolt	24
M4 x 35 bolt	4
M4 x 45 bolt	14
1/2-13 x 1-1/2 in bolt + nut, greased (forms the dowel hole)	2
3/8-16 x 2-1/2 in bolt, greased (holds the top plate)	4
5/8 in plain steel rod, greased (forms a tie-rod hole), cut to 140 mm, 243 mm, 377 mm, 377 mm	2

Tools: a drill with sharp brad-point bits (sizes in section 5) and a 2-1/2 in hole saw; a jigsaw or a circular saw for the upright side plates' sloped edge; a square and a long straightedge; a few clamps; a 7 mm nut driver and a 3 mm hex key; a marker that writes on melamine.

3. Print the parts

42 prints, 0.31 kg of PETG and about 24 hours of printing, sliced in Bambu Studio for an A1 mini (settings in the print guide). Print one corner bracket and one saddle first and check them on an offcut of your board: the saddle should drop over it with a little play.

Folder	File	Qty	g (all)	Time each
upright	`kit-back-bracket.stl`	4	21	27m 33s
upright	`kit-front-bracket.stl`	4	32	38m 27s

Folder	File	Qty	g (all)	Time each
upright	`kit-saddle-flat.stl`	4	18	22m 40s
upright	`filler-toe-filler.stl`	2	54	1h 33m 58s
upright	`former-foot-gasket-groove-part1of4.stl`	2	7	17m 8s
upright	`former-foot-gasket-groove-part2of4.stl`	2	3	8m 46s
upright	`former-foot-gasket-groove-part3of4.stl`	2	3	8m 46s
upright	`former-foot-gasket-groove-part4of4.stl`	2	7	17m 10s
upright	`plug-slope-plug-lower.stl`	2	19	45m 49s
upright	`plug-slope-plug-upper.stl`	2	19	45m 47s
upright	`plug-tie-rod-pocket.stl`	4	33	39m 54s
upright	`kit-plate-spacer.stl`	4	15	20m 9s
upright	`kit-post-bushing.stl`	4	68	1h 11m 44s
upright	`cast-in-post-l1-plug.stl`	4	14	15m 49s

- **Nuts:** press M4 nuts into the hex pockets of the brackets and saddles; slide the nuts into the pockets of the formers and fillers (a dab of grease keeps them from falling out). A nut on a bolt through the hole pulls a tight one into place.
- **Pegs:** test-fit the parts of each long former; a light pass with a file if a peg is tight.

4. Cut the boards

Most boards are strips, so the quickest way is to have the store rip each sheet into strips at the widths below, and crosscut them at home. Label every board with its name on the outside face as you cut it, and mark which face is the cast face (the melamine face that the mix touches).

Board	Qty	Size (mm)	Notes
Upright floor	2	198 x 575	
Upright side plate, outer	2	495 x 271 (profile)	cut to the profile
Upright side plate, inner	2	495 x 271 (profile)	cut to the profile
Upright back board	2	80 x 271	
Upright front board	2	80 x 312	ends cut square
Upright bridge	2	131 x 108	
Upright cleat, outer	2	405 x 40	offcut strip, flat on the floor
Upright cleat, inner	2	405 x 40	offcut strip, flat on the floor
Upright cleat, back	2	80 x 40	offcut strip, flat on the floor

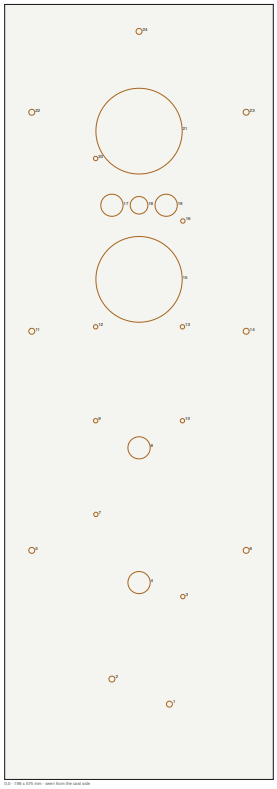
The upright side plates are cut to a profile (their drawings give the corners): cut the rectangle, then the sloped front edge. The front board's ends are square.

5. Drill the boards

- Drill every hole from the **cast side**, so any tear-out is on the outside. Clamp a scrap under the board where the bit comes out.
- The drawings and tables are seen from the cast side: x from the left edge, y from the bottom edge. For a wall the bottom edge sits on the floor; for a floor, the bottom of the drawing is toward the front wall (the beam's bottom wall). Left and right are as you look at the cast face, so a floor is mirrored compared with the machine. Bridges and cleats are drawn from the top.
- The brackets and saddles are drill guides: clamp one in place and drill through its holes.
- A washer goes under every bolt head; countersink nothing.

Hole	Drill	For
3.2 mm	1/8 in	foot-gasket-groove
4.5 mm	11/64 in	cleat, corner bracket, saddle, slope-plug-lower, slope-plug-upper, toe-filler
10 mm	25/64 in	3/8-16 bolt into the top plate
13.1 mm	33/64 in	1/2-13 bolt, base dowel hole
16.3 mm	41/64 in	5/8 core rod
63.5 mm	2-1/2 in hole saw	hole saw

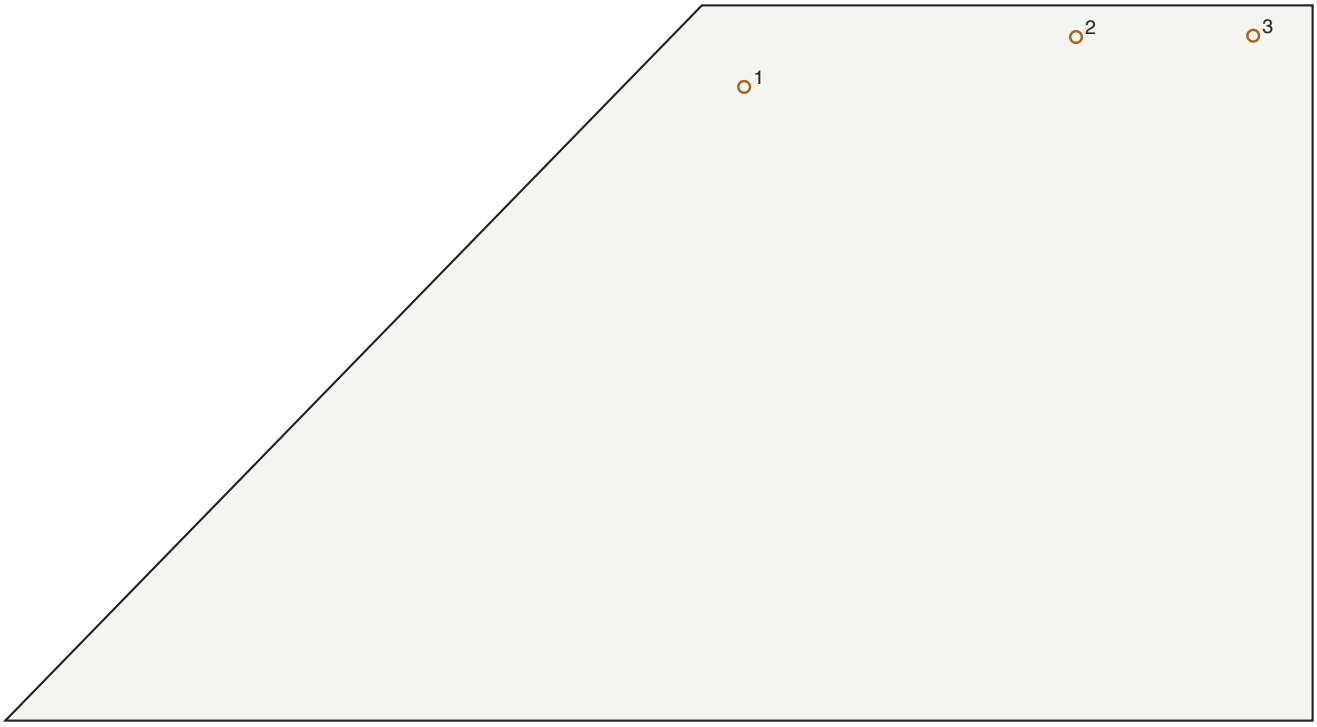
Upright floor (x2)



#	x	y	Hole	For
1	121	56	4.5	toe-filler

#	x	y	Hole	For
2	79	75	4.5	toe-filler
3	131	136	3.2	foot-gasket-groove
4	99	146	16.3	5/8 core rod
5	20	170	4.5	cleat
6	178	170	4.5	cleat
7	67	197	3.2	foot-gasket-groove
8	99	246	16.3	5/8 core rod
9	67	266	3.2	foot-gasket-groove
10	131	266	3.2	foot-gasket-groove
11	20	333	4.5	cleat
12	67	336	3.2	foot-gasket-groove
13	131	336	3.2	foot-gasket-groove
14	178	333	4.5	cleat
15	99	371	63.5	hole saw
16	131	414	3.2	foot-gasket-groove
17	79	426	16.3	5/8 core rod
18	99	426	13.1	1/2-13 bolt, base dowel hole
19	119	426	16.3	5/8 core rod
20	67	461	3.2	foot-gasket-groove
21	99	481	63.5	hole saw
22	20	495	4.5	cleat
23	178	495	4.5	cleat
24	99	555	4.5	cleat

Upright side plate, outer (x2)

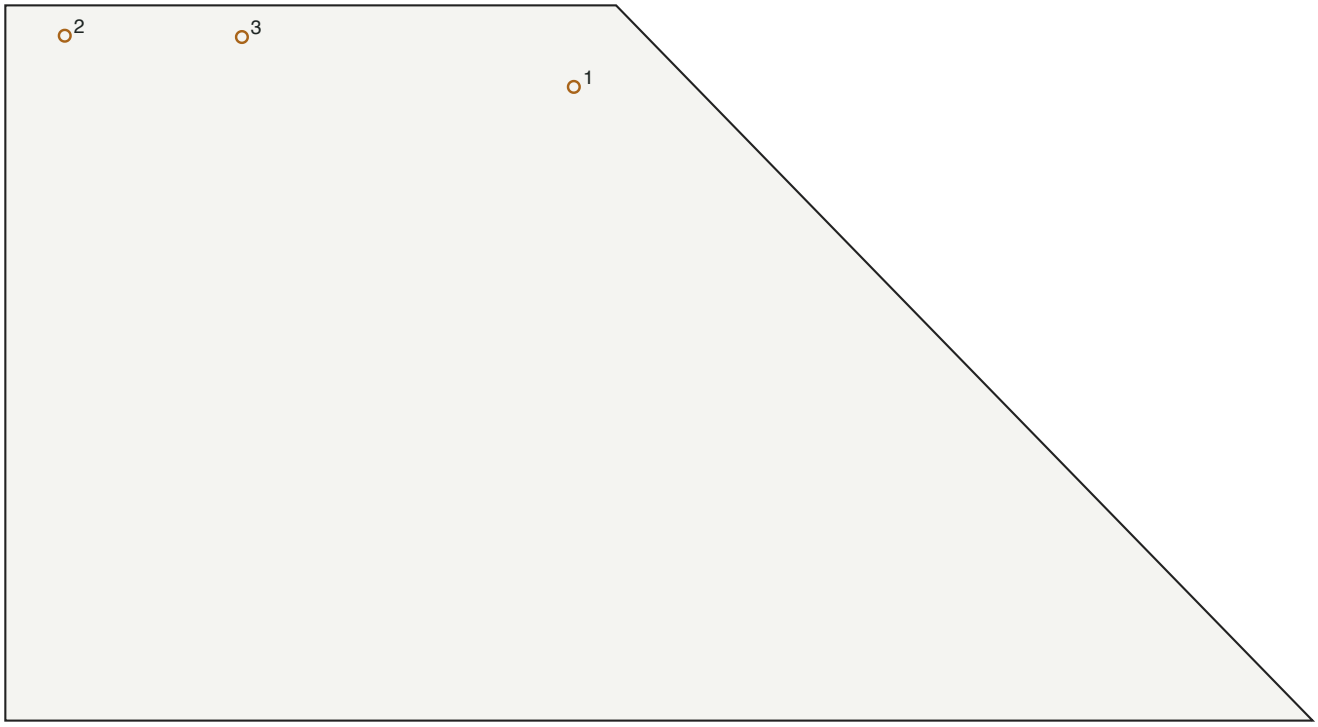


0,0 · 495 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	280	240	4.5	corner bracket
2	406	259	4.5	saddle
3	473	260	4.5	corner bracket

Outline corners (x, y): (0, 0), (495, 0), (495, 271), (264, 271)

Upright side plate, inner (x2)

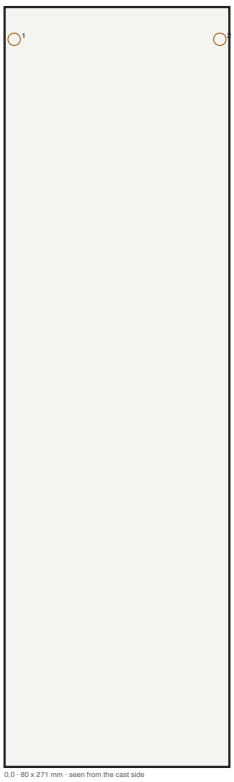


0,0 · 495 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	215	240	4.5	corner bracket
2	22	260	4.5	corner bracket
3	90	259	4.5	saddle

Outline corners (x, y): (495, 0), (0, 0), (0, 271), (231, 271)

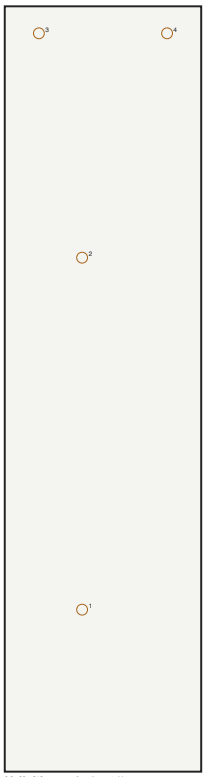
Upright back board (x2)



0,0 - 80 x 271 mm - seen from the cast side

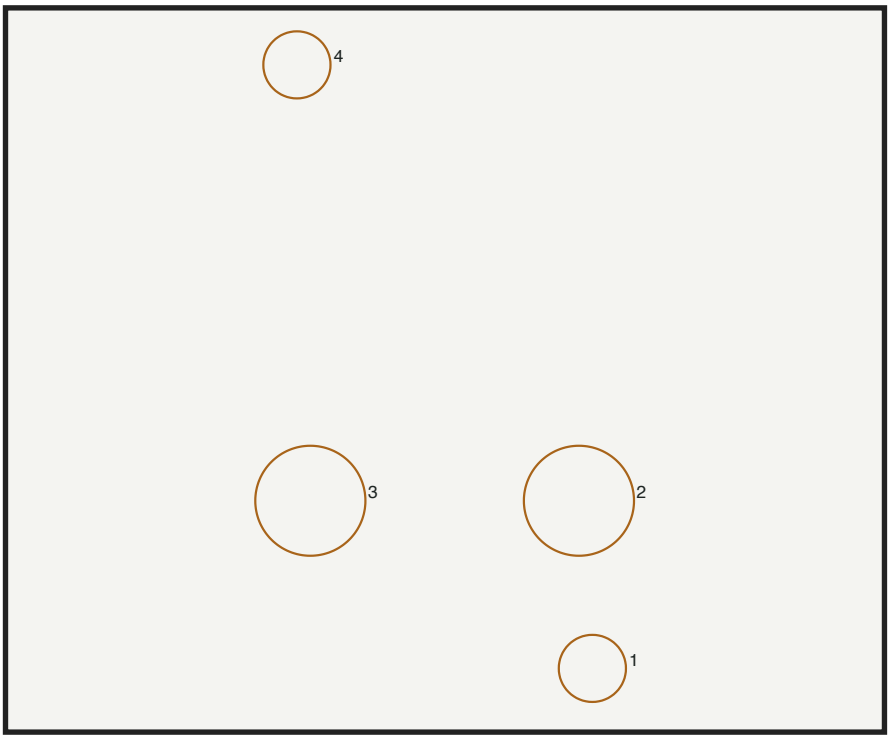
#	x	y	Hole	For
1	3	260	4.5	corner bracket
2	77	260	4.5	corner bracket

Upright front board (x2)



#	x	y	Hole	For
1	32	66	4.5	slope-plug-lower
2	32	209	4.5	slope-plug-upper
3	14	301	4.5	corner bracket
4	66	301	4.5	corner bracket

Upright bridge (x2)



0,0 · 131 x 108 mm · seen from the outside (top)

#	x	y	Hole	For
1	87	10	10	3/8-16 bolt into the top plate
2	85	34	16.3	5/8 core rod
3	45	34	16.3	5/8 core rod
4	43	100	10	3/8-16 bolt into the top plate

Upright cleat, outer (x2)



0,0 · 405 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	365	20	4.5	cleat
2	203	20	4.5	cleat
3	40	20	4.5	cleat

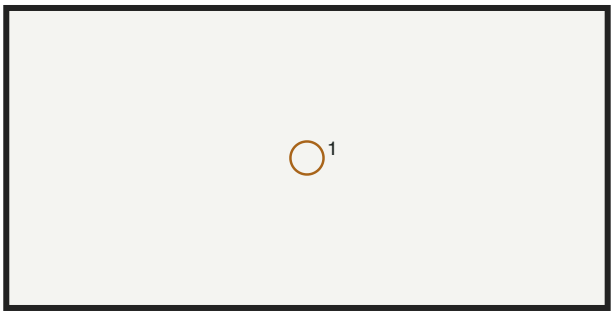
Upright cleat, inner (x2)



0,0 · 405 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	365	20	4.5	cleat
2	203	20	4.5	cleat
3	40	20	4.5	cleat

Upright cleat, back (x2)



0,0 · 80 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	40	20	4.5	cleat

6. Prepare the surfaces

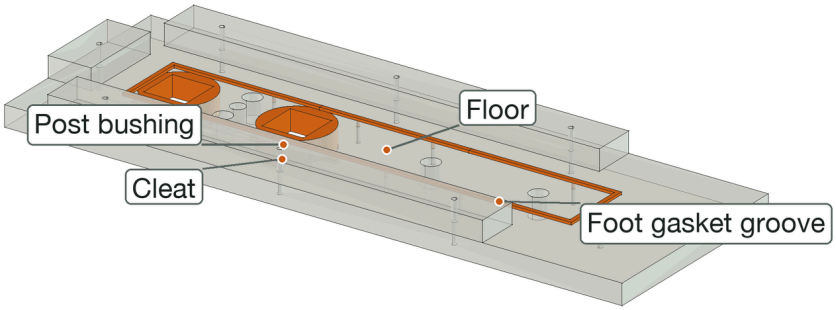
- Wax every face the mix touches, boards and removable printed pieces alike: 3-5 thin coats of paste wax, each buffed, then a coat of PVA. Don't wax the pieces that stay in (core posts, cast-in parts): scuff them and wipe them with acetone so the epoxy grips.
- The castings' outside edges are rounded 10 mm: glue 3/8 in quarter-round molding into the inside corners that form outside edges (floor to wall), and run a small silicone fillet into the others.
- Grease every screw, bolt and rod that will come out of the casting.

7. Put the mold together

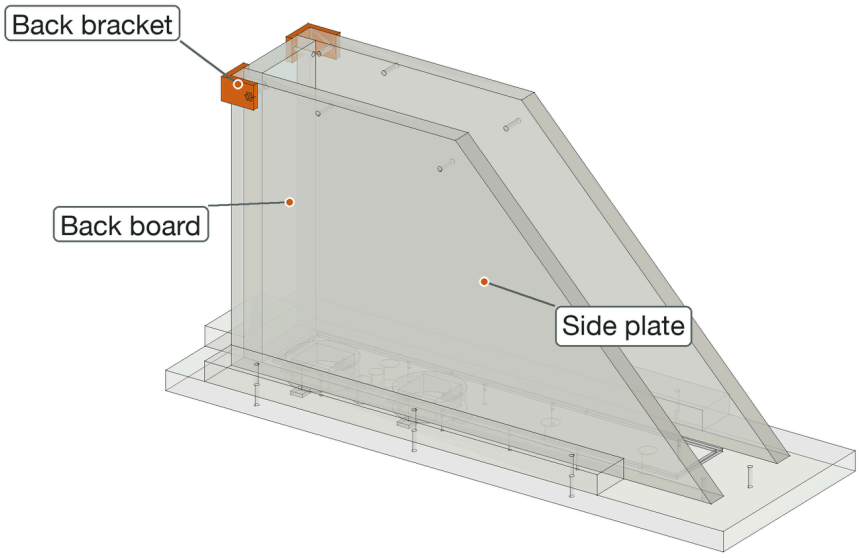
Uprights (two molds, cast together)

The uprights are cast foot down. Both are the same, so build two of everything.

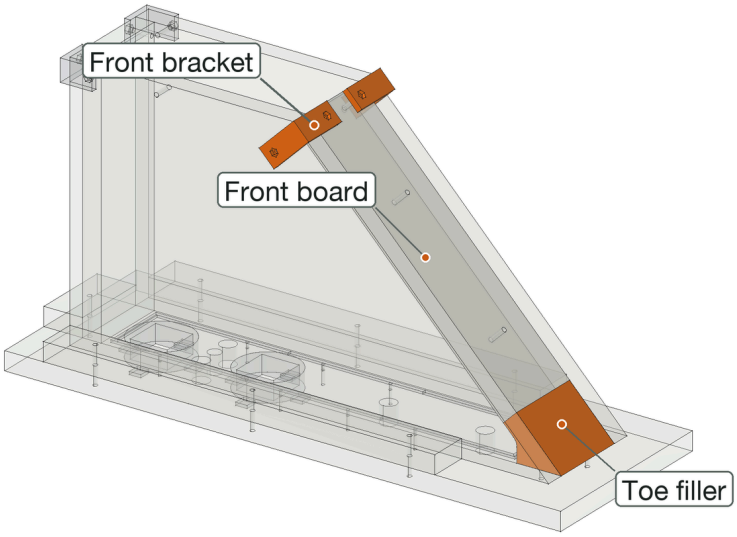
1. Push the post bushings into the floor from below and pin the gasket-groove bead to it. Bolt the cleats down.



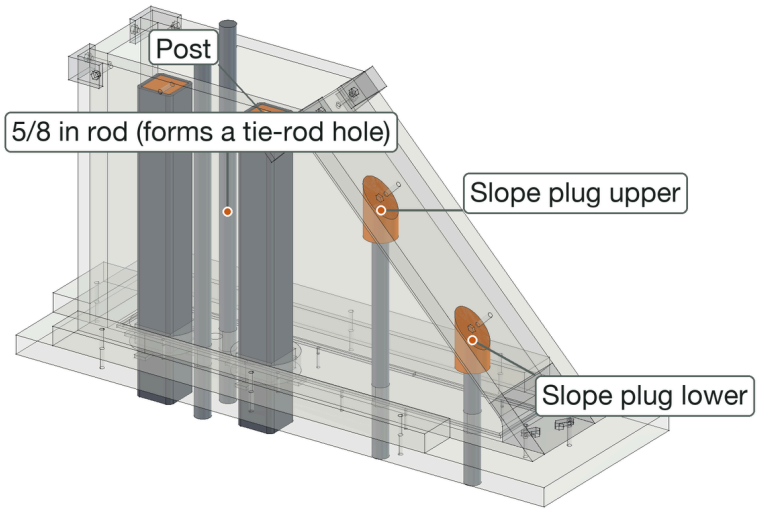
2. Stand the side plates against their cleats and the back board between them; fit the two back brackets.



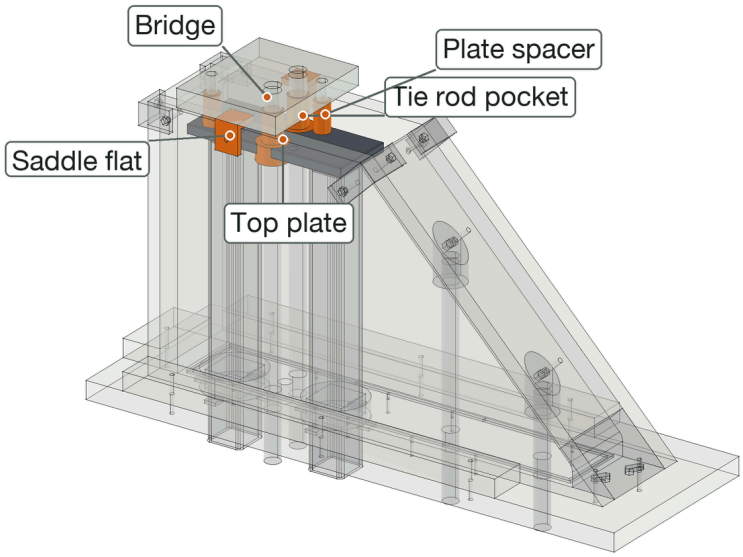
3. Bolt the toe filler to the floor between the side plates, set the front board's foot in its lip and fit the two front brackets at the top.

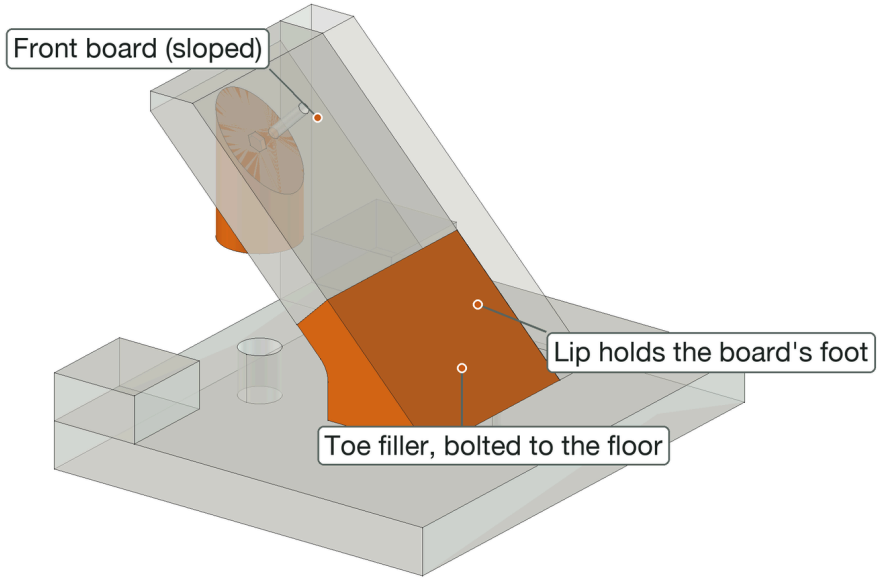


4. Stand the mold on its four 2x4 blocks. Push the posts up through the bushings onto the bench, and the greased 5/8 in rods up through the floor: the two short ones into the slope plugs (bolted to the front board), the two long ones on through the top. The dowel-hole bolt and nut go in from underneath.



5. Fill to the line and vibrate. Then slide the top pocket plugs down the long rods, hang the top plate from the bridge on its greased 3/8 bolts and spacers, and drop the bridge onto its saddles.





8. Pouring, and the pour line

- The pour line is 25 mm below the wall tops. Mark it inside each wall with a pencil line, and screed the top with a straight board that has two 25 mm feet riding on the wall tops.

- Vibrate each layer until it stops bubbling; the brackets and cleats hold the walls, so you can vibrate the floor board hard. Check the corner brackets and saddle bolts are still tight halfway through.
- Keep the mix off the bridges and saddles; scrape it off before it cures.

9. Taking it apart, and reuse

1. Leave it the full cure time (build guide). Unbolt the bridges and saddles, then the corner brackets, then the cleats; pull the walls straight off (a plastic wedge in the corner helps).
2. Unscrew every insert screw, coupler bolt and rod from outside, then lift the casting off the floor (or the floor off the casting).
3. Unbolt and pull the formers and fillers (the bolts come out from the board side first). A wooden dowel and a light tap frees a stuck one.
4. Scrape the boards, re-wax them, and store the kit pieces with their nuts in a box per mold. The same kit does the next casting; only the formers and pieces that stay in are per casting.

10. If something doesn't fit

Problem	Fix
A nut won't slide under the hole	Clean the pocket with a nut on a bolt; a warm nut (hair dryer) pushes in
A bracket or saddle is loose on the board	Your board is thinner than set: shim with tape, or set `mold_board` and reprint
A pin won't go into its hole	Run a 1/8 in bit through the hole again; trim the pin tip with a knife
A foam core moves while pouring	Check its post bolt is tight and the post is glued; add carpet tape under it
Mix leaks at a corner	A silicone fillet inside the corner before waxing; check the diagonals and brackets
A former is stuck after curing	Bolts out first; then a dowel and a light tap from its thick end